SIZE	DWG NO	SH	REV
A	328T7632	1	-

THIRD ANGLE PROJECTION

REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
	Initial Release	23-04-2026	Muzammil Hasan Siddiqui

THIS DRAWING TO BE REVISED IN MICROSOFT WORD ONLY. THE SOURCE FILE IS STORED IN THE NEXUS CONTROLS DMS

LIST OF COMPLEMENTARY DOCUMENTS		
ITEM	**DESCRIPTION**	**DWG. NO.**
4108	Network Topology	148M1528
A014	Wiring Diagram	325T0860
A014	System Layout	324T4549

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES ON:	SIGNATURES	DATE
2 PL DECIMALS ±	DRAWN Gali Rama Prasad	2026-04-23
	CHECKED Muzammil Hasan Siddiqui	2026-04-23
3 PL DECIMALS ±	ENGINEER Gali Rama Prasad	2026-04-23
ANGLES ±	ISSUED SAP PLM	
FRACTIONS ± ✓		
APPLIED PRACTICES		
ET: 1	SIMILAR TO: NONE	

GE VERNOVA

FACTORY ACCEPTANCE TEST PLAN

FIRST MADE FOR:3BR-PS1000037 MLI 0203

SIZE	CAGE CODE	DWG NO
A	NONE	328T7632
SCALE:	NONE	SHEET 1 of 20

TABLE OF CONTENTS

1. INTRODUCTION

This document describes the proposed Factory Acceptance Test (FAT) of the Mark VIe turbine control panel and HMI equipment.

The Factory Acceptance Test will take place in Jafza FZE during a period of two days. Testing will include hardware, software, and a documentation review of the system. The intent of the test is to demonstrate that the system under test operates as designed and meets all requirements of the contract specifications. This document will be used as the approval document to provide a record of the test activities.

At the conclusion of the FAT, a summary of the test results and corrective actions will be added to this document and archived. Any items that cannot be resolved prior to shipment will be addressed by engineering and communicated to the site installation team prior to the unit outage by the most effective means possible.

2. REQUISITION INFORMATION

Customer:	SAUDI ELECTRICITY CAMPANY
End User:	JIZZAN POWER PLANT
Site Name:	SAUDI ARABIA-JIZZAN POWER PLANT
Equipment Number:	297873, 298248
Equipment Model:	7EA
Driven Equipment:	GENERATOR
SAP Project #:	5RY-PS6300016
GE Personnel:	Surya Narayana Rao Chinta (Project Manager) Muzammil Hasan Siddiqui (Project Engineer) Gali Rama Prasad (Functional Engineer) Yousif Malik (Functional Engineer) Arnel Maderazo (Functional Engineer)

3. AGENDA

Day 1 – 9:00AM to 4:00PM

- Introduction
- Review Drawings, Hardware Inspection & I/O testing
- Lunch
- Communication Testing
- Daily review (Points of Interest, Points of Note, Change Log)

Day 2 – 9:00AM to 4:00PM

- Review of any changes implemented
- Complete Software Functional Testing
- Lunch
- Complete Hardware & I/O Testing
- Wrap-up and sign off

4. PRE-FAT TESTING

Prior to the FAT, the control panel is inspected against the engineering drawings for proper components, power distribution, grounding, and other internal wiring. After power on, the power supplies and internal distribution is rung out and the panel is inspected for any hardware faults. IONet communication and controller functionality are verified during the first download of the site software.

An Advanced Dry Rig (ADR) is used to simulate the various I/Os to the turbine control panel, including digital signals, 4-20mA analog inputs, magnetic pickups, servo outputs, LVDTs, and others. Once connected, the ADR is used to check proper functionality of all I/O points in the panel, including spares. The simulation software used to drive the ADR is developed alongside the site turbine software and used to validate the site software via desktop simulation and during pre-FAT functional testing using the ADR test stand and site HMIs.

5. FACTORY ACCEPTANCE FUNCTIONAL TESTING

The following sections provide a brief description of the inspection / test, permits the recording of comments and allows sign off of each particular test.

During this FAT the turbine control panel will be connected to a dry-rig capable of simulating many of the unit operating parameters as well as exercising the actual I/O via hard-wired connections. Simulation code will be integrated within the dry-rig simulator in order to verify overall functionality of the controller(s), the HMI computers, and communications between various system components.

As testing is completed, any agreed residual actions should be recorded in Section 6, and/or in a separate residual action tracking document to be attached at the conclusion of the test.

5.1 HARDWARE INSPECTION

(Note: Repeat these tests for each control cabinet, as needed)

Step	Description	Comments and Observations
5.1.1.	**Control Cabinet Inspection** Inspect cabinet for complete and proper installation of the equipment. Inspect metalwork and paintwork. Log any missing equipment along with general comments.	
	Date: __________ Customer Approval(s)________________	
5.1.2.	**Network and HMI Hardware Inspection** Inspect all network and HMI hardware including HMI computers, monitors, printers, peripherals, network switches, time servers, and media converters. Log any missing equipment along with general comments.	
	Date: __________ Customer Approval(s)________________	

Additional Comments:

5.2 INPUT/OUTPUT TESTING

(Note: Repeat these tests for each control cabinet, as needed)

Step	Description	Comments and Observations
5.2.1.	**Discrete Input Testing** Agree on which discrete inputs to simulate for test. When it is not possible to test all points, then 20% of points shall be tested on a random basis. Verify each point by monitoring the HMI display, or the ToolboxST display. Associated diagnostic alarms are to be observed.	
	Date: __________ Customer Approval(s)_______________	
5.2.2.	**Discrete Output Testing** Agree on which discrete outputs to simulate for test. When it is not possible to test all points, then 20% of points shall be tested on a random basis. Verify each point by monitoring the test stand and HMI display. Associated diagnostic alarms are to be observed.	
	Date: __________ Customer Approval(s)_______________	
5.2.3.	**Analog Input Testing** Agree on which analog inputs to simulate for test. When it is not possible to test all points, then 20% of points shall be tested on a random basis. Verify each point by monitoring the HMI display, or the ToolboxST display. Associated diagnostic alarms are to be observed.	
	Date: __________ Customer Approval(s)_______________	
5.2.4.	**Analog Output Testing** Agree on which analog outputs to simulate for test. When it is not possible to test all points, then 20% of points shall be tested on a random basis. Verify each point by monitoring the HMI display, or the ToolboxST display. Associated diagnostic alarms are to be observed.	
	Date: __________ Customer Approval(s)_______________	

SIZE	DWG NO	SH	REV
A	328T7632	8	-

Additional Comments:

GE VERNOVA	SIZE	CAGE CODE	DWG NO	
DRAWN GALI RAMA PRASAD	A	NONE	328T7632	
ISSUED 2026-04-23	SCALE:	NONE		SHEET 8 of 20

Step	Description	Comments and Observations
5.2.5.	**RTD Input Testing** Agree on which RTD inputs to simulate for test. When it is not possible to test all points, then 20% of points shall be tested on a random basis. Verify each point by monitoring the HMI display, or the ToolboxST display. Associated diagnostic alarms are to be observed.	
	Date: __________ Customer Approval(s)______________	
5.2.6.	**Thermocouple Input Testing** Agree on which TC inputs to simulate for test. When it is not possible to test all points, then 20% of points shall be tested on a random basis. Verify each point by monitoring the HMI display, or the ToolboxST display. Associated diagnostic alarms are to be observed.	
	Date: __________ Customer Approval(s)______________	
5.2.7.	**Vibration Input Testing** Agree on which Vibration inputs to simulate for test. When it is not possible to test all points, then 20% of points shall be tested on a random basis. Verify each point by monitoring the HMI display, or the ToolboxST display. Associated diagnostic alarms are to be observed.	
	Date: __________ Customer Approval(s)______________	
5.2.8.	**Actuator Driver Testing** All actuator (4-20mA and servo) outputs are to be functionally tested. Verify each point by monitoring the HMI display, or the ToolboxST display. Associated diagnostic alarms are to be observed.	
	Date: __________ Customer Approval(s)______________	
5.2.9.	**Magnetic Pickup Input Testing** Frequency proportional to turbine speed is to be injected for both control and protection speed inputs. Verify each point by monitoring the HMI display, or the ToolboxST display. Perform 2oo3 validation for TMR inputs. Associated diagnostic alarms are to be observed.	
	Date: __________ Customer Approval(s)______________	

Additional Comments:

5.3 REDUNDANCY TESTING

(Note: Repeat these tests for each control cabinet, as needed)

Step	Description	Comments and Observations
5.3.1.	**TMR Redundancy/CPU Power Loss** Cycle power to each of the cores (R, S, T), IONet switches, and other equipment as necessary. Confirm that in the case of controllers and IONet switches, failure of one device will not trip the running turbine, and correct alarms are given.	
	Date: __________ Customer Approval(s)_____________	
5.3.2.	**Network Redundancy** Confirm that in the case of network switches failure of one device will not trip the running turbine and correct alarms are given. Remove power to all network switches and verify the running turbine does not trip.	
	Date: __________ Customer Approval(s)_____________	

Additional Comments:

5.4 COMMUNICATION TESTING

Step	Description	Comments and Observations
5.4.1.	**Controller Download Test** Demonstrate an online download software to the turbine control panel through ToolboxST. Confirm ability to download program and configuration files to the control panel without tripping the running turbine.	
	Date: __________ Customer Approval(s)______________	
5.4.2.	**Trip Log Verification** Confirm proper functionality of the trip log recorder by simulating a turbine trip. Open and review the trip log trend to verify proper configuration.	
	Date: __________ Customer Approval(s)______________	
5.4.3.	**External Communication Test** As applicable, configure a test PC which emulates a Modbus, GSM, or OPC client and connect to the control network. Demonstrate that data passes to and from the control panel to the HMI and external client by observing the external client data as equivalent control panel data changes.	
	Date: __________ Customer Approval(s)______________	

Additional Comments:

5.5 HMI SCREEN TESTING

Step	Description	Comments and Observations
5.5.1.	**HMI Screen Review and Testing** Navigate through each HMI screen. Verfiy the look and feel is consistent, and that color animations are correct. Test all navigation and commands to verify functionality. This test can be completed in conjuction with other functional testing.	
	Date: __________ Customer Approval(s)_____________	
5.5.2.	**HMI Screen Configuration Verification** Confirm the units of measure are set to English or Metric as required. Demonstrate language selection capability and accuracy as applicable.	
	Date: __________ Customer Approval(s)_____________	

Additional Comments:

5.6 TURBINE CONTROL SOFTWARE TESTING

Step	Description	Comments and Observations
5.6.1.	**Startup Verification** Verify the turbine is in ready-to-start state. Select start and allow the test setup to simulate a startup. Verify the start, acceleration rates, and note any abnormal alarms or trips. Verify operator interface functionality.	
	Date: __________ Customer Approval(s)_______________	
5.6.2.	**Shutdown Verification** Select stop while the turbine is running at full speed and allow the test setup to simulate a fired shutdown. Verify ability to interrupt shutdowns prior to falling below operating speed. Verify the unit flames out at the proper speed. Note any abnormal alarms or trips. Verify any active trip does not allow start or restart.	
	Date: __________ Customer Approval(s)_______________	
5.6.3.	**Generator Synchronization Verification** Exercise auto, manual, and monitor sync modes as applicable. Confirm all modes operate correctly. Exercise excitation raise/lower and speed/load raise/lower via manual sync. In all modes, verify HMI display and operator interface functionality.	
	Date: __________ Customer Approval(s)_______________	
5.6.4.	**Speed Control Functionality** Exercise the speed governor control. Confirm controller functionality, HMI display functionality. Confirm auto load controller modulates HP shaft speed setpoint. Confirm HP shaft speed remains stable at full speed no load, and a selection of loaded points up to temperature control.	

Step	Description	Comments and Observations
5.6.5.	**Fuel Control Functionality – Liquid Fuel** Select Liquid Fuel for startup. During startup, FSNL, and selected loaded conditions, confirm that the fuel regulator calls for the proper fuel schedule, and is stable at all steady state conditions. Verify fuel transfer at loaded conditions as applicable. Date: __________ Customer Approval(s)__________	
5.6.6.	**Temperature Control Functionality** Exercise the speed governor control. Confirm that fuel system calls for fuel reduction to match exhaust temperature schedule.	
5.6.7.	Date: __________ Customer Approval(s)______________	
5.6.8.	**IGV Control Functionality** Exercise the IGV control modes. Verify during startup and loading of the unit, the modulated IGV control opens or closes the guide vanes according to the minimum setpoints and temperature schedule. Verify manual operation of the IGV, and manual IGV temperature control as applicable.	
5.6.9.	**Full Load Rejection Test** Load the unit and open the generator breaker. Verify that the full load rejection software keeps flame in the unit and drops to FSNL.	
	Date: __________ Customer Approval(s)______________	

Additional Comments:

5.7 TURBINE PROTECTION SOFTWARE TESTING

Step	Description	Comments and Observations
5.7.1.	**Overspeed Protection Test** Select the overspeed test mode from the HMI display, and raise the turbine speed to the trip setpoint. Verify the turbine trips at the expected speed and record the speed (RPM) in the comments. Verify proper alarms are annunciated. Repeat as applicable for electrical and mechanical tests. Verify that restart is not permitted without master reset.	
	Date: __________ Customer Approval(s)______________	
5.7.2.	**Overtemperature Protection Test** Simulate overtemperature conditions. Verify the turbine trips on high temperature, and that proper alarms are annunciated. Verify that restart is not permitted without master reset.	
	Date: __________ Customer Approval(s)______________	
5.7.3.	**High Temperature Spread Protection Test** Simulate temperature spread conditions. If applicable, verify first failed high thermocouple alarm annunciates and no trip occurs. Simulate high temperature spread and verify turbine trips, and proper alarms are annunciated. Verify that restart is not permitted without master reset.	
	Date: __________ Customer Approval(s)______________	

5.7.4.	**Protection Core Firmware Trip Tests** Verify that protection card firmware trips are functional per specification and alarms are generated. Suggested tests and verification methods are as follows: • Control speed signal trouble (L12HFD_C) • Protective speed signal trouble (L12HFD_P) • Deceleration trip (L12H_ACC): all shafts – Force dry rig speed to 0 when turbine is running. • Contact trips (where applicable): Force dry rig wired input to open. • E-Stop (L5ESTOP1): Activate the E-stop pushbutton. • Speed Differential trip (L4SPDDT): Force dry rig speed to different value at control and protection inputs. If speed inputs are wired from same dry rig output, then force Speed1 variable (Variables tab) to a different value (disable stale speed trip before test) when turbine is running. See trip help to calculate speed difference value to activate trip. • Stale Speed Trip (L4SSPDT): Force "Speed1" variable (on the Variables tab) to current value when turbine is running. • Watchdog Trip (L4CWDT): Force ContWdog variable to current value.	
	Date: __________ Customer Approval(s)_____________	

Additional Comments:

5.8 OPEN TESTING AND COMMENTS

GE Vernova Approval	Date: _________	Initials________
Customer Approval	Date: _________	Initials________

6. AGREED RESIDUAL ACTIONS

Description:

Identify residual actions, questions, agreements, or any other issues to be addressed after the FAT but before installation starts. Include who will be doing the work and the estimated completion date. Additional residual action lists may be attached at the end of this document as needed.

Item #	Description of Change or Action	Action Item Owner	Estimated Completion Date
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			

GE Vernova Approval	Date: __________	Initials________
Customer Approval	Date: __________	Initials________

7. FACTORY ACCEPTANCE TEST SIGNATURE SHEET

The undersigned agree that the factory acceptance test was completed on the date indicated.

GE Vernova *Approval*	CUSTOMER *Approval*
______________________________ (signature)	______________________________ (signature)
____________ (date)	____________ (date)

GE Vernova Controls

FACTORY ACCEPTANCE TEST

The following approvals constitute the satisfactory completion of the Factory Acceptance Test (FAT).

The FAT was conducted on a structured basis as outlined in the previous pages.

The above signatures constitute an agreement on the outstanding residual actions and a commitment by GE Vernova Controls to resolve and/or complete any outstanding action items. Outstanding items are identified on the Agreed Residual Action Items list or as an attachment to this document.

Customer's acceptance authorizes the billing associated with the Factory Acceptance Test completion.